

MUHAMMAD HUZAIFA AFTAB

Pakistan's Youngest SOLIDWORKS Expert & Champion, Mechanical Engineering Undergraduate,
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PROFESSIONAL SUMMARY

Mechanical Engineering undergraduate at UET Taxila Pakistan's Youngest Certified SOLIDWORKS Expert (CSWE) and SOLIDWORKS Champion, recognised globally by Dassault Systèmes. Founder of MechMate Hub, a mentoring platform that has guided 50+ engineers internationally to CAD/CAE certification success. Experienced in FEA, CFD, and design validation of complex systems including jet engines, UAVs, wind turbines, and heat exchangers. Active researcher in smart materials and 4D printing, with industrial exposure through a six-week internship at Heavy Mechanical Complex (HMC) Taxila. Aspire Leaders Program graduate with a record of entrepreneurship, leadership, and global collaboration driven to reshape technical education access for engineers in the developing world.

KEY ACHIEVEMENTS

Youngest SOLIDWORKS Certified Expert (CSWE) & SOLIDWORKS Champion in Pakistan recognised by Dassault Systèmes globally.

20+ SOLIDWORKS certifications earned, including CSWP, CSWA, Simulation, FEA, Advanced Drawing, and more.

50+ Engineers mentored internationally to CAD/CAE certification success through MechMate Hub.

Top 10 Performer Aspire Pakistan Founders Edge Cohort III (entrepreneurship & innovation programme).

Harvard Aspire Leaders Program Graduate global leadership initiative.

5 Depts Internship coverage at HMC Taxila: Production Planning, Mechanical Works, Foundry & Forge, QA/QC, Technology.

CORE COMPETENCIES

- SOLIDWORKS (CSWE) 3D Modelling, Assemblies, 2D Drawings
- Finite Element Analysis (FEA)
- Computational Fluid Dynamics (CFD)
- Product Design & Concept Development
- Prototyping & Design Validation
- Manufacturing: Foundry, Forging, Machining, QA/QC
- Project Management Lean Six Sigma (White/Yellow/Green Belt)
- Google PM Certificate
- Design Documentation Engineering Specs, Reports, Technical Manuals
- Primavera P6

PROFESSIONAL EXPERIENCE

SOLIDWORKS Champion, Dassault Systèmes

February 2026 – Present

- Selected for a global network recognized by Dassault Systèmes for technical excellence in 3D CAD and engineering workflows.
- Recognized as the youngest SOLIDWORKS Champion from Pakistan, with experience in product design, 3D modeling, and assembly management.
- Shared knowledge, best practices, and mentorship with designers, students, and engineering professionals worldwide, helping them improve their CAD and engineering workflows.

Founder & Lead Mentor, MechMate Hub

August 2024 – Present

- Founded and scaled a technical mentoring platform focused on CAD/CAE learning, supporting engineering students globally.
- Mentored 50+ learners internationally, guiding them to SOLIDWORKS certification success through structured preparation pathways.
- Designed and delivered curriculum covering 3D modelling, FEA, CFD, and advanced SOLIDWORKS tools directly relevant to design validation and simulation workflows.
- Built structured certification preparation frameworks adopted by learners across multiple countries.

Summer Intern, Heavy Mechanical Complex (HMC), Taxila

July 2025 – August 2025

- Completed a six-week industrial internship across five departments: Production Planning & Control, Mechanical Works, Foundry & Forge Works, Quality Assurance/Control, and Technology (FFW).
- Gained hands-on exposure to large-scale manufacturing processes including metal casting, forging, machining, and component assembly directly informing design-for-manufacture decision-making.
- Applied quality inspection principles and industrial workflow analysis consistent with precision engineering environments.

Research Intern, MultiOmics

December 2025 – Present

- Contributing to a review article on smart materials and additive manufacturing (4D printing), strengthening technical writing and analytical research skills.
- Independently identified and proposed the review paper topic smart materials, shape memory alloys, and 4D printing to the research group; proposal accepted and under development.
- Performing structured literature review and scientific analysis under expert mentorship, developing systematic research methodology aligned with academic publication standards.

Facilitator & Intern, Nobel Learning PBC, Remote

August 2025 – Present

- Conduct professional sessions, manage stakeholder engagement, collect feedback, and support program delivery.
- Participated in leadership development, collaborative learning, and professional growth activities, contributing to team coordination and program engagement.

EDUCATION

BSc in Mechanical Engineering | University of Engineering & Technology (UET) Taxila

August 2023 – August 2027 (Expected)

- Active member of ASME UET Taxila Chapter Assistant Vice Chair, Media & Graphics.

RESEARCH INTERESTS

Smart Materials & Shape Memory Alloys. 4D Printing & Additive Manufacturing. Computational Simulation. Finite Element Analysis. Advanced Manufacturing. Mechanical Design. Sustainable Engineering Systems.

CERTIFICATIONS & CREDENTIALS

- Certified SOLIDWORKS Expert (CSWE) | highest-level SOLIDWORKS certification, globally recognised
- 20+ SOLIDWORKS Certifications | CSWP, CSWA, Simulation, FEA, Advanced Drawing, and more
- Certified Istec Vibration Training Category 1+
- Google Project Management Professional Certificate (v2)

- Lean Six Sigma | White, Yellow & Green Belt
- LEAP Platform Sustainable Energy Modelling (SDPI)

LEADERSHIP & PROFESSIONAL MEMBERSHIPS

- Affiliate Member | Institution of Mechanical Engineers (IMechE)
 - Student Member | American Society of Mechanical Engineers (ASME)
 - Member | International Association of Engineers (IAENG)
 - Assistant Vice Chair (Media & Graphics) | ASME UET Taxila Chapter
 - Team Lead | UMEED-E-SUBH UET Taxila (Welfare Society)
 - Aspire Leaders Program Graduate | Harvard-affiliated global leadership initiative
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